

KBeaconPro App Instruction

Revision History

Version	Date	Change Description	Author
V1.0	2017/11/12	Initial draft for KBeaconTools	Adam
V1.2	2018/02/23	Name updating	Elaine
V1.3	2022/11/2	Adding Power Profiler function	Elaine
V1.4	2024/08/28	Add S7 and some other feature	Elaine

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1. Download KBeaconPro App

Download the App 'KBeaconPro' from iOS App Store or Android Google Play or scan the QR code below to down the App.



iOS APP



Android APP

Minimum requirements

A mobile phone with Bluetooth 4.0 support is needed. For Android devices, Android version 5.1 or newer. For iOS devices, iOS version 10.0 or newer.

Reminder: This instruction uses Android App to demonstrate. The iOS App interface is slightly different from Android App.

2. How to Connect KBeacon Device to KBeaconPro App

Kindly note: Please make sure your KBeacon device is with battery already.

Enable your smart phone Bluetooth and run the App 'KBeaconPro'.

2.1 Turn on KBeacon

● KBeacon with external or internal button (Including model

K1/K11/K21/K23/K3/K4/K41/K4P/K5/K51/K5P/K6/K61/K6P/K7/K71/K8/K81/W59/F1/W3/W52/B1/B2/S1/C2/S3/S4)

The factory setting of KBeacon with button is power OFF.

- ✧ Turn ON the device: Hold the button for 3 seconds, the LED starts flashing and the flashing lasts for 30 seconds.
- ✧ Turn OFF the device: Hold the button for 5 seconds, the LED starts flashing and flash 8 times.

● KBeacon without button (Including model K12/K15/K15A/K15L/W4/U1/K9/K91/K9P/W7/S2/P1)

The factory setting of KBeacon without button is ON. K12/K15/K9/P1/W7 does not support turning off the power.

● KBeacon without button (Including model T1)

It's powered off in default. Use the magnet to activate/turn on the device.

2.2 KBeacon capability matrix

 K1/K11	 K21/K23	 K12	 K3
 K4/K41/K4P	 K5/51/K5p	 K6p/K6pt/K6pb	 K7/K71
 K8/K81	 K9/K9p/K91	 W4	 W59
 U1	 F1	 W51	 W52
 W3	 W7	 B1	 B2
 S1	 S2	 P1	 W8
 K15/K15A/K15L	 C2	 K16	 T1

2.2.1 Trigger Capability

Model	Trigger type	Trigger action
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		Advertisement	Record	Report to App	Alarm
K11	Button(single click/double/triple/long press)	yes	no	yes	yes
K15a	Motion	yes	no	yes	no
K16a	Motion	yes	no	yes	no
K21	Motion	yes	no	yes	yes
	Button(single click/double/triple/long press)	yes	no	yes	yes
K3	Motion	yes	no	yes	no
K4/K4p	Motion	yes	no	yes	no
K4pt	Motion	yes	no	yes	no
	humidity above trigger humidity below trigger	yes	yes	yes	no
K5/K5p	Motion	yes	no	yes	no
K5pt	Motion	yes	no	yes	no
	humidity above trigger humidity below trigger	yes	yes	yes	no
K6p/K6pb	Motion	yes	no	yes	no
	humidity above trigger humidity below trigger	yes	yes	yes	no
	temperature above trigger temperature below trigger	yes	yes	yes	no
K6pt	Motion	yes	no	yes	no
	temperature above trigger temperature below trigger	yes	yes	yes	no
S1	Cutoff trigger(Door open/close)	yes	yes	yes	no
S2	PIR trigger	yes	yes	yes	no
S3	PIR trigger	yes	yes	yes	no
	light level above/below trigger	yes	yes	yes	no
	Motion	yes	no	yes	no

S4	Cutoff trigger(water trigger)	yes	no	yes	no
S5	Motion	yes	no	yes	no
	humidity above trigger humidity below trigger	yes	yes	yes	no
	temperature above trigger temperature below trigger	yes	yes	yes	no
W9	Motion	yes	no	yes	yes
	Button(single click/double/triple/long press)	yes	no	yes	yes
B1/B2/B3	Button(single click/double/triple/long press)	yes	no	yes	no
P1	Motion	yes	no	yes	no

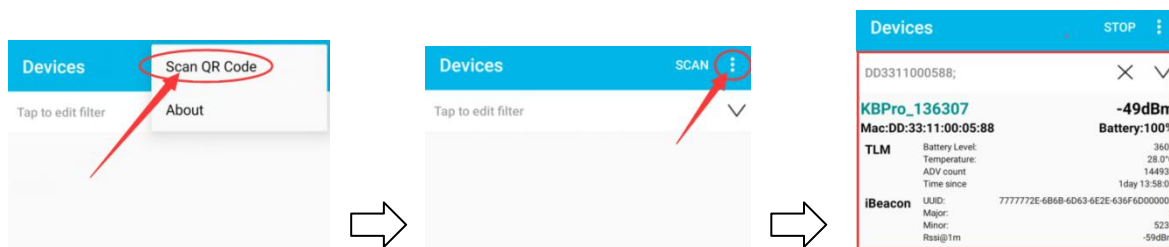
2.3 Connect KBeacon

Let's use a K9P (MAC: DD3311000588) to demonstrate, the MAC ID is printed on the device:



● Method 1: Scan QR code to connect

Run KBeaconPro App, Find 'Scan QR code' on the App, and then scan the QR code on the device to find this KBeacon device quickly.



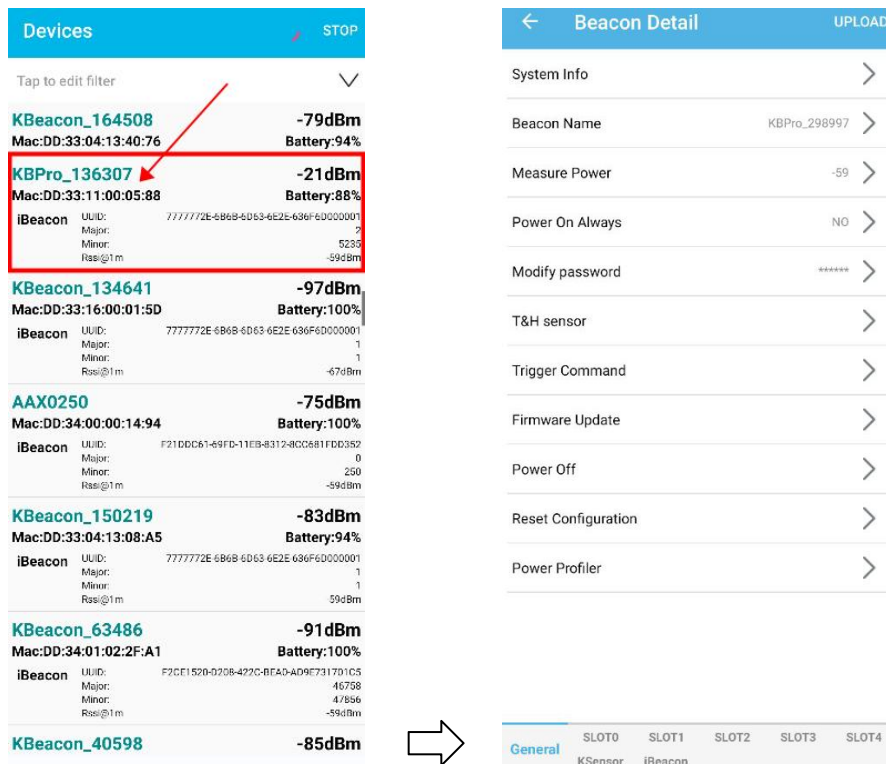
Reminder: iOS App filter the device by Device Name when use the 'Scan QR code' method. If the device name is not KBPro, the device can not be found on iOS App when scan the QR code.

● Method 2: Scan the Bluetooth signal to connect

Run KBeaconPro App and tap 'SCAN' in the top right corner, the App can scan the device's Bluetooth signal, then

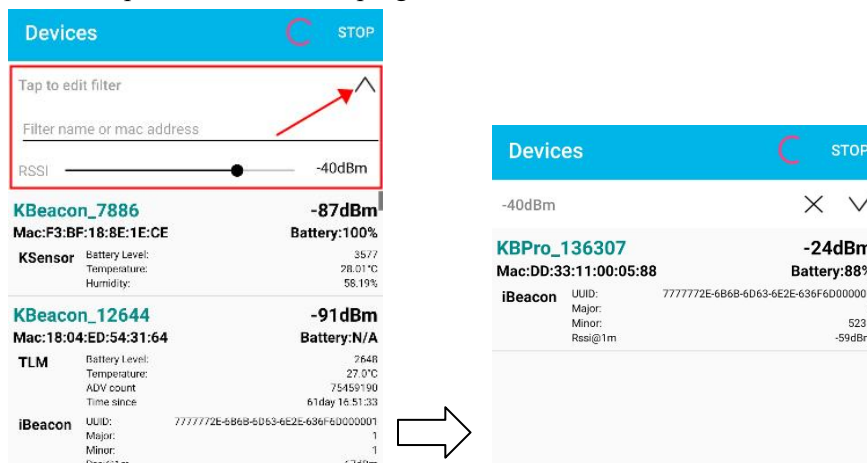
the Beacon device will be displayed on the scan page.

Find the corresponding KBeacon device on the App according to its MAC ID, Tap it, it will start connecting and jump to the configuration page (see pictures below).



If there are too many devices found , filter by RSSI to find a certain Beacon quickly.

Put the KBeacon device close to your phone (within 10cm range). Slide the RSSI bar to set the RSSI value at -30~-40dBm, tap the arrow on the top right corner, then the nearest KBeacon can be found



3. How to Configure KBeacon

3.1 SLOT Definition

KBeacon supports total 5 SLOTS (SLOT0 to SLOT4). Each SLOT is independent and configurable. The Beacon

type can be set for each SLOT. Beacon parameters such as Adv interval, Tx Power, Connectable enable/disable etc can be configured separately for each SLOT. They are independent of each other.

Each SLOT can be set to one Beacon type ONLY. For example, if you set SLOT0 to be iBeacon, set SLOT1 to be URL, set SLOT2 to be TLM, then the KBeacon device will broadcast iBeacon, URL and TLM simultaneously.

KBeacon supports the following Beacon types:

- **iBeacon**
- **Eddystone URL**
- **Eddystone UID**
- **Eddystone TLM**
- **KSensor:** KKM self-defined protocol, includes battery level and sensor information
- **System:** Including the KBeacon device info such as System ID (ie.MAC ID), Model name.
- **EBeacon:** Encrypt the broadcast, Beacon will encrypt the UUID with AES, and will change the AES key according to the UTC time. This kind of broadcasting prevents cracking and copying.

Note: Some Beacons do not support KSensor and EBeacon broadcasts, please refer to the Beacon Datasheet for details.

3.2 How to configure iBeacon and Eddystone

Take iBeacon as an example:

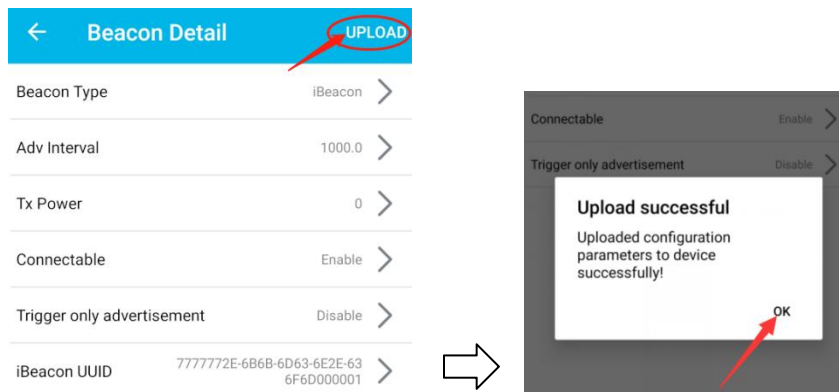
Tap: Beacon Type—> iBeacon—> Save—> Return

iBeacon parameters (UUID, Major ID, Minor ID, Adv Interval, TX power etc.) can also be configured in the App. Eddystone URL, UID, TLM, can be configured by the same steps above.

Parameters	Defaults	Describe
Adv Type	iBeacon	Disable: Slot does not broadcast KSensor: KKM sensor data, see Section 7.3 for details UID/TLM/URL: Google Eddystone, see Section 7.2 for details iBeacon: Apple iBeacon, , see Section 7.1 for details System: KKM System data, see Section 7.5 for details System: KKM EBeacon data, see Section 7.4 for details
Adv Interval	1000.0	Advertisement period is expressed in decimal and the unit is ms. If you need to use it on Apple devices, it is recommended to follow Apple's specifications. Apple has some suggestions that make the device more easily discovered by IOS phones. (The suggest value was: 152.5 ms; 211.25 ms; 318.75 ms; 417.5 ms; 546.25 ms; 760 ms; 852.5 ms; 1022.5 ms; 1285 ms). For more information, please refer to Section 3.5 in "Bluetooth Accessory Design Guidelines for Apple Products". https://developer.apple.com/accessories/Accessory-Design-Guidelines.pdf .
Adv Channel	0~6	By default, Beacon broadcasts on channels 37, 38, and 39. You can also set this slot to broadcast on only one or two channels.
Tx power	-40 ~ 8?	Beacon TX power. The value range depends on the support

		capability of the device, some devices are -40~4dBm, some devices are -40~8dBm.
Connectable	Yes	Whether the device can be connected. For detailed description, please refer to Section 6.1.
Trigger only advertisement	No	When this feature is set to be 'YES', this slot will be broadcasted only when the trigger happens. For example, if you set 'Trigger Adv Slot' of 'Button single click' to be SLOT0 and SLOT0 is iBeacon, then iBeacon will be broadcasted only when the button single click happens.
iBeacon parameters (UUID/Major/Minor)	NA	Configuration parameters of iBeacon

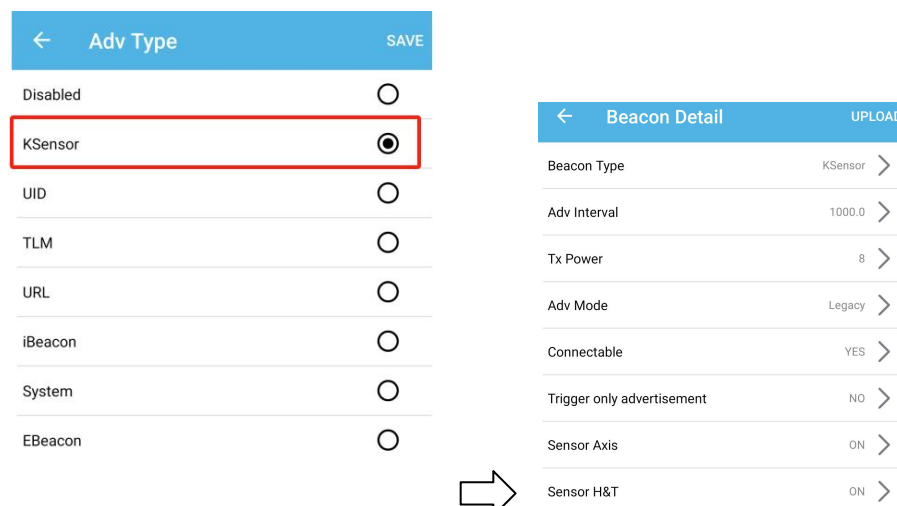
After the parameters are modified, you need to tap: UPLOAD—> OK, then all the parameters configured can be loaded to the device successfully.



3.3 How to configure KSensor and System

KSensor is KKM defined protocol, it includes battery level and sensor information (for example temperature&humidity sensor, acceleration sensor etc).

Tap: Beacon Type—> KSensor—> Save—> Return



Parameters	Defaults	Describe
Sensor Axis	Yes	When KBeacon supports accelerometer sensor, whether to include 3-axis information in KSensor broadcasts.
Sensor H&T	Yes	When KBeacon supports temperature and humidity sensor, whether to include temperature and humidity information in KSensor broadcasts.
Sensor Light	Yes	When KBeacon supports light sensor, whether to include light level in KSensor broadcasts.
Sensor PIR	Yes	When KBeacon supports light sensor, whether to include PIR events in KSensor broadcasts.
Sensor VOC	Yes	When KBeacon supports VOC sensor, whether to include VOC in KSensor broadcasts.
Sensor CO2	Yes	When KBeacon supports CO2 sensor, whether to include CO2 in KSensor broadcasts.
New Log Count	Yes	When KBeacon supports record sensor history, whether to include New Log Count in KSensor broadcasts.

“System” can also be configured by the same steps above.

3.4 How to configure EBeacon

EBeacon is a broadcast format defined by KKM. This broadcast format can encrypt the 16-bit UUID with AES before broadcasting, preventing third parties from obtaining the plain UUID. In addition, in order to prevent the broadcast from being cloned, Beacon will dynamically change the secret key based on UTC time. In this way, third parties cannot clone the device through the broadcast content.

[Reminder: Only some Beacons support EBeacon encrypted broadcasting. For details, please refer to the corresponding Beacon datasheet.](#)

← Adv Type
SAVE

Disabled	☐
KSensor	☐
UID	☐
TLM	☐
URL	☐
iBeacon	☐
System	☐
EBeacon	☒

← Beacon Detail
UPLOAD

Adv Type	EBeacon	>
Adv Interval	1000.0	>
Adv Channel	7	>
Adv Mode	Legacy(BLE 4.0)	>
Tx Power	8	>
Connectable	YES	>
Trigger Only Adv	NO	>
UUID	7777772E-6B6B-6D63-6E2E-636F6D000001	>
AES Update Interval	5	>
AES Type	AES ECB	>

➡

Parameters	Defaults	Describe
UID	Yes	The UUID that needs to be encrypted for broadcast
AES Update Interval	5	AES KEY update interval, unit is second. To ensure that the device is not cloned, Beacon will update the AES KEY at a certain interval. The calculation method of AES KEY is related to UTC time.
AES Type	ECB	UUID encryption algorithm, the default is AES ECB encryption.

3.5 General parameters

Beacon has some common parameters, such as device name, device password, calibration power, etc. You can configure this in the general page.

In addition, we may need to issue some commands to the device. For example, a factory reset.

3.5.1 General parameters

← Beacon Detail
UPLOAD

System Info	>
Beacon Name	KBPro_476719 >
Measured Power	-59 >
System Info	>
Power On Always	NO >
Modify Password	***** >

Parameters	Defaults	Describe
Beacon Name	KBPro_xx	Beacon includes the device name in the Bluetooth broadcast. The default device name is KBPro_xx and the xx is the device ID. For regular products, this xx will be printed in a QR code sticker. Example: The device names of the following Beacons are: KBPro_476719 
Measure Power	-59	The received signal strength at 1 meter when the Beacon TX power is 0dBm. This parameter is used for iBeacon and Eddystone protocols. Example 1: If SLOT0 broadcasts the iBeacon protocol and the transmission power is 4dBm, then the Power (RSSI@1m) in the iBeacon message = -59dBm + 4dBm = -55dBm Example 2: If SLOT0 broadcasts the Eddystone UID and the transmission power is 4dBm, then the Power (RSSI@0m) in the UID message = -18dBm + 4dBm = -14dBm
Power On Always	False	This parameter is valid for devices with buttons. To prevent the device from being shut down by long pressing the button, we can set

		this parameter to TRUE. When this parameter is set to TRUE, the long press button power off function will be disabled.
Password		The default password of the device is 16 characters 0000000000000000. When the APP connects to this device, Beacon will verify the password. If the verification fails, the connection will not be allowed. Warning: If you change the password, please remember the password, otherwise the app will no longer be able to connect again.

3.5.2 Power off

For KBeacon device WITH BUTTON, you can use the App to turn off the device.

Tap: Power Off—> OK

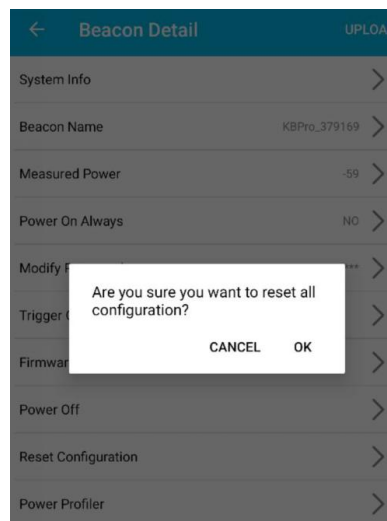
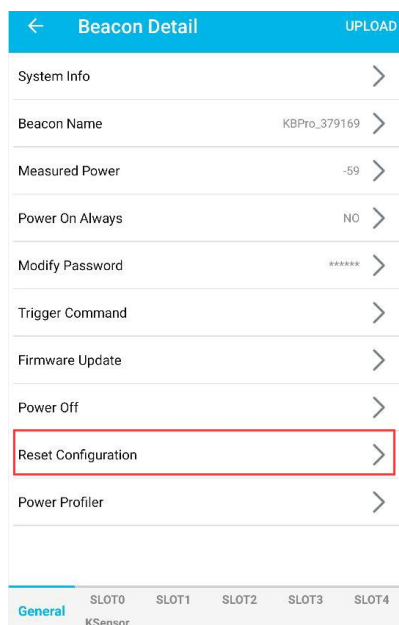
After powering off, the device enters a low-power state. Beacon will stop broadcasting. If the Beacon has a sensor, it will stop the sensor data collection.

3.5.3 Reset configuration

You can reset the KBeacons setting to default configuration on the App.

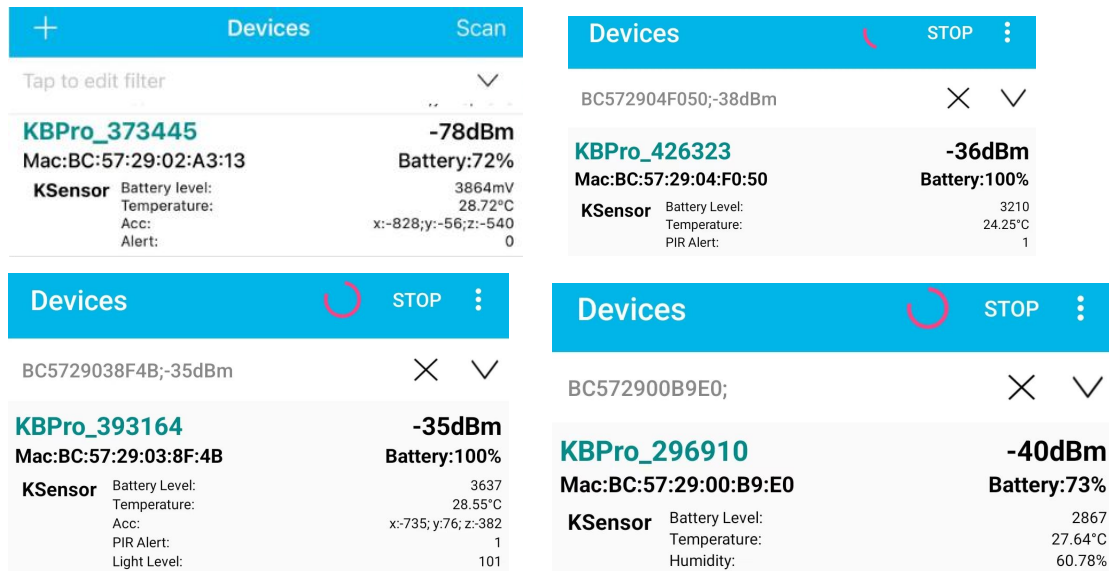
Reminder: If you customize some of your own parameters to KKM, such as Trigger parameters, multiple slots broadcast parameters. KKM will configure these parameters for you before leaving the factory. If you perform the "reset configuration" operation, these pre-configured parameters may be lost.

Tap: Reset configuration—> OK



3.6 View KSensor advertisement data

When we configure KSensor to broadcast sensor information in section 3.3, we can scan sensor data through app.



The broadcast content of different sensors is displayed differently on the APP.

Type	Describe
Battery Level	The unit of battery level is mV. For example, if it is 3210, it means the battery voltage is 3210mV.
Temperature	The unit of temperature is °C. For example, if it is 25°C, it means that the current temperature detected by the beacon is 25°C.
Humidity	The unit of humidity is %. For example, if it is 60%, it means that the current humidity detected by the beacon is 60%.
Acc	Acc means the acceleration sensor. It includes the value of Axis X, Axis Y and Axis Z, and the unit is mg.

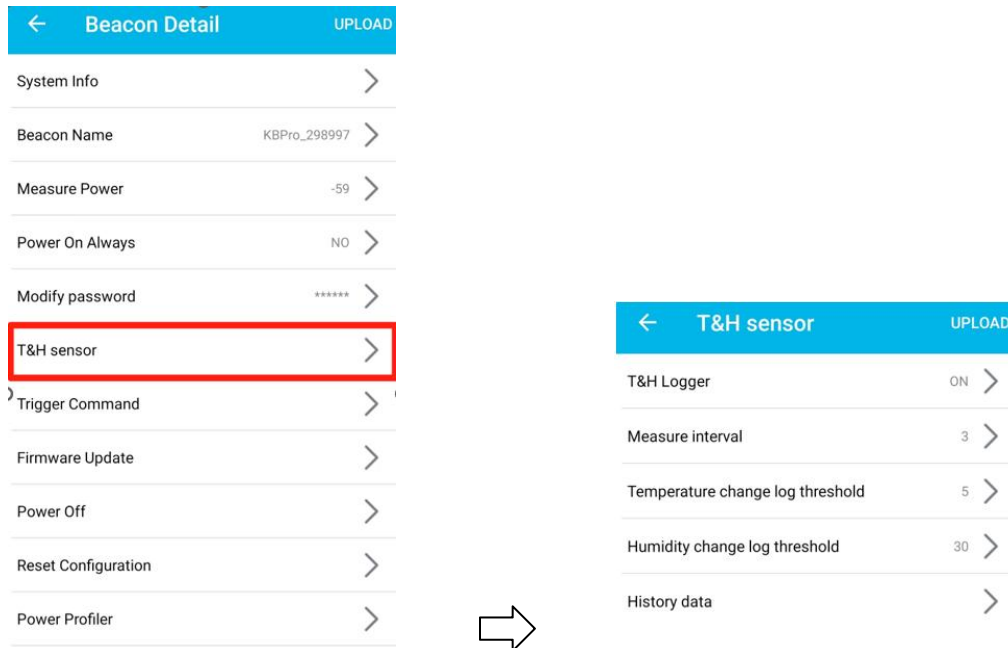
Alert	For door sensor(S1): ● 0: no alert ● 1: door was open ● 2: KBeacon was unplug ● 3: KBeacon was unplug and door was open For cutoff sensor(W7/W3): ● 0: no alert ● 1: watchband was cutoff For water sensor (S4): ● 0: no detected water ● 4: detected water For car sensor(S7): ● 0: no parking ● 1: Parking was detected ● 64: Beacon's geomagnetic sensor is not calibrated ● 32: Beacon does not have a reference geomagnetic value set ● 96: Beacon is not calibrated and there is no reference geomagnetic value
PIR Alert	0: no PIR alert 1: PIR alert
Light Level	The unit of light level is lux. Value range: 1 ~ 65535 For example, if it is 101, it means that the current light level detected by the beacon is 101 lux.
New log count	The number of newly generated records since the last time the App or gateway read the records. The App or gateway can decide to connect devices and read new records based on the New log count.

3.7 How to configure sensor parameters

For some KBeacon models with sensor. We need to configure some sensor parameters. Such as the interval of sensor detection. Whether to record events to Flash. Whether the detection is not performed at a specific time, such as the sensor enters a sleep state at night.

- Temperature & Humidity Sensor (K23/K6P/K6PS/K6PB/K6PT)
- PIR sensor(S2)
- VOC and CO2 sensor

3.7.1 Temperature & Humidity Sensor



Type	Describe
T&H Logger	When this feature is on, the device will record when temperature/humidity change exceeds the follow threshold you set.
Temperature change log threshold	If you set a value (take 5 as an example, it means 0.5 degrees Celsius), the device will save the record if the difference between the current temperature and the last saved temperature greater or equal than 0.5 degrees Celsius.
Humidity change log threshold	If you set a value (take 20 as an example, it means 2%), the device will save the record if the difference between the current humidity and the last saved humidity greater or equal than 2%.
Measure interval	Time interval for measuring temperature and humidity.
History data	Tap to view history data

Reminder: If the Temperature change log threshold is set to 0, a temperature and humidity record is recorded for each measurement interval.

History data: The data can be loaded or cleared.

← T&H history		EXPORT	⋮
2022-06-16 18:09:19	Temperature: 29.86°C Humidity: 70.31%		
2022-06-16 18:07:16	Temperature: 30.05°C Humidity: 67.04%		
2022-06-16 18:04:19	Temperature: 30.2°C Humidity: 70.14%		
2022-06-16 17:55:31	Temperature: 29.79°C Humidity: 71.45%		
2022-06-16 17:54:19	Temperature: 29.5°C Humidity: 74.47%		
2022-06-16 17:53:22	Temperature: 29.05°C Humidity: 74.84%		
2022-06-16 17:51:31	Temperature: 29.19°C Humidity: 71.2%		
2022-06-16 17:49:31	Temperature: 29.66°C Humidity: 69.17%		



← T&H histo		Load All	Clear
2022-06-16 17:53:22	Temperature: 29.05°C Humidity: 74.84%		
2022-06-16 17:51:31	Temperature: 29.19°C Humidity: 71.2%		
2022-06-16 17:49:31	Temperature: 29.66°C Humidity: 69.17%		
2022-06-16 17:38:39	Temperature: 29.89°C Humidity: 72.2%		
2022-06-16 17:24:45	Temperature: 29.65°C Humidity: 69.19%		
2022-06-16 17:21:39	Temperature: 29.77°C Humidity: 72.11%		
2022-06-16 17:16:39	Temperature: 29.63°C Humidity: 69.15%		
2022-06-16 17:01:15	Temperature: 29.31°C Humidity: 72.18%		

3.7.2 Door/PIR sensor

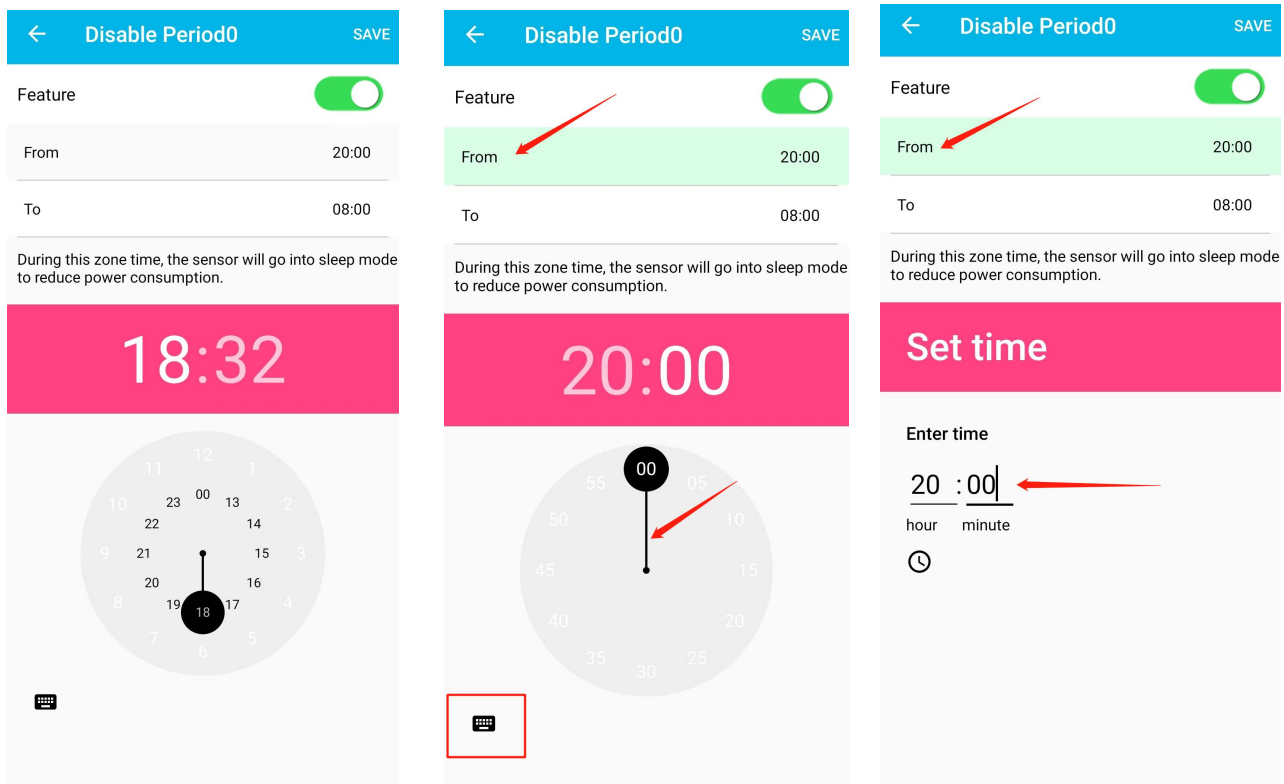
For door sensor, you can configure when the sensor works.

← Beacon Detail		UPLOAD
System Info	>	
Beacon Name	KBPro_299037 >	
Measure Power	-59 >	
Power On Always	NO >	
Modify password	***** >	
Cutoff sensor	>	
Trigger Command	>	



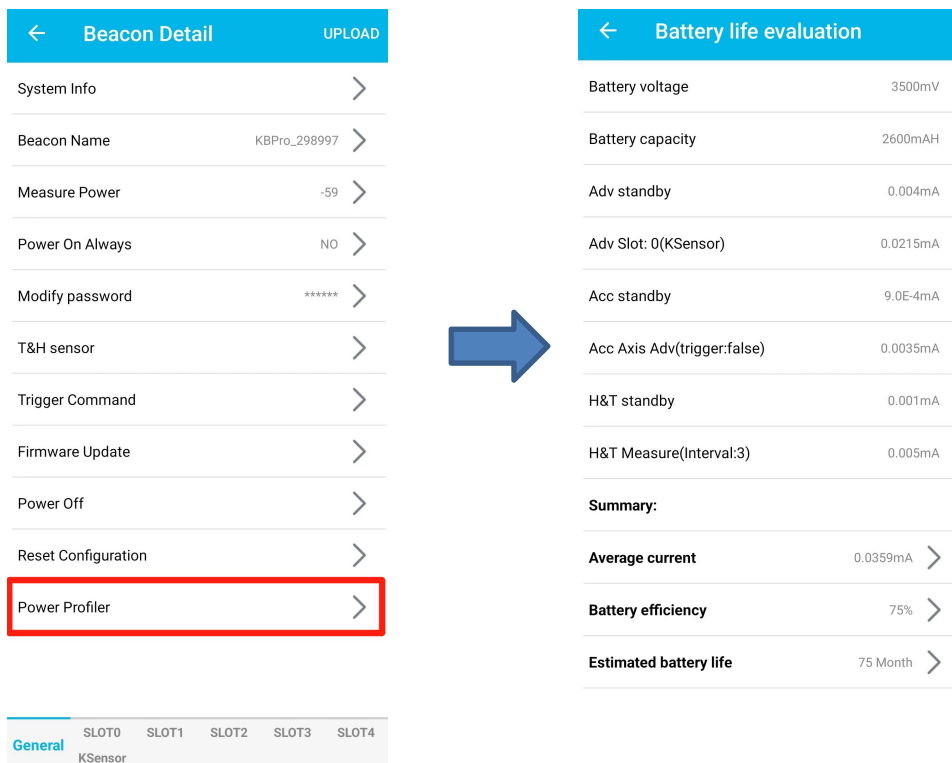
← Cutoff sensor		UPLOAD
Disable Period0	00:00 ~ 00:00 >	
Disable Period1	00:00 ~ 00:00 >	
Disable Period2	00:00 ~ 00:00 >	
History data	>	

Type	Describe
Disable Period0/1/2	When this feature is on, you can set the time period for sleep mode by turning the hour hand and minute hand, or input the time through the keyboard.



3.8 How to evaluate battery life

KBeaconPro supports evaluating the battery life of beacon according to the configured parameters.



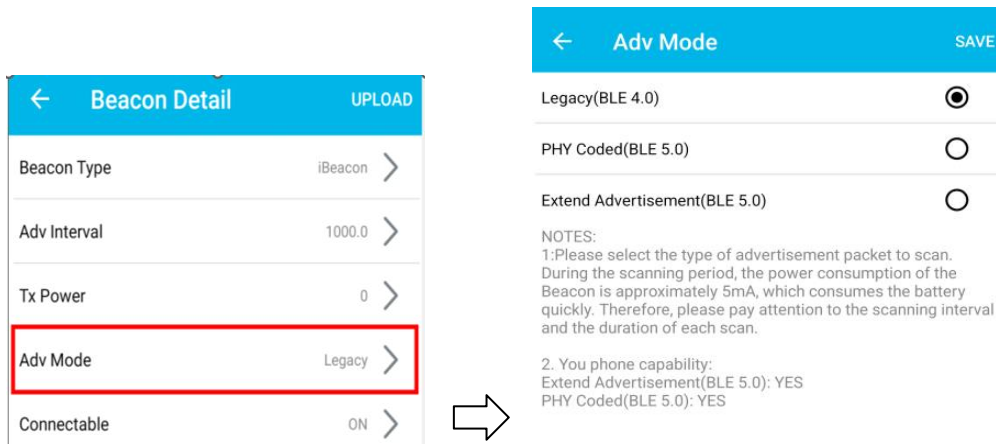
Type	Describe
Adv standby	When the device is in standby mode, it also consumes a certain amount of power. This power consumption is usually between 1~4uA.
Adv slot: 0	When a slot broadcast is enabled, the slot will periodically send broadcast messages, which will generate a certain power consumption.
Acc standby	When the device has an acceleration sensor, even if the acceleration sensor is not working, there will be about 0.9uA of power consumption.
Acc Axis Adv	If you set KSensor to broadcast 3-axis information, the accelerator will start measuring. If the broadcast interval is shorter, the power consumption will be higher.
H&T standby	When the device has an humidity sensor, even if the sensor is not working, there is about 1uA of power consumption.
H&T measure (Interval:3)	Indicates the power consumption when the temperature and humidity sensor measures once every 3 seconds. The shorter the measurement interval, the higher the power consumption.
Average current	The average current of the device is based on the current configuration parameters, and calculated after the device is powered on for 30 seconds for current stabilizes. The average current does not include power consumption by trigger broadcasting. Also, it does not include the power consumption when the device is connected.
Battery efficiency	Usually the battery capacity is based on the ideal 1mA discharge model at room temperature. In actual use, the capacity of the battery is related to temperature, current and self-discharge. We recommend 75%.
Estimated battery life	= Battery capacity * Battery efficiency/ Average current/24(hours)/30(days)

3.9 How to configure advertisement mode

For some KBeacon models that support BLE5.0 long range feature, 'Adv Mode' can be configured.

- **Legacy:** BLE 4.0 advertisement
- **PHY Code:** BLE 5.0 long range feature
- **Extend Advertisement :** BLE5.0 Extend broadcasting

[KBeaconPro App can detect which Adv Mode your phone support \(Only supported on Android phones\).](#)



Reminder:

Please make sure that your phone supports BLE 5.0 PHY Code (Long range) feature, otherwise you will not be able to scan the PHY code advertisement if the Beacon was set to PHY Code (Long range) Mode.

If you set the Beacon to PHY code advertisement, and your phone doesn't support PHY Code broadcast, you can force the device to enter the Legacy mode for 30 seconds by single click the button of the device.

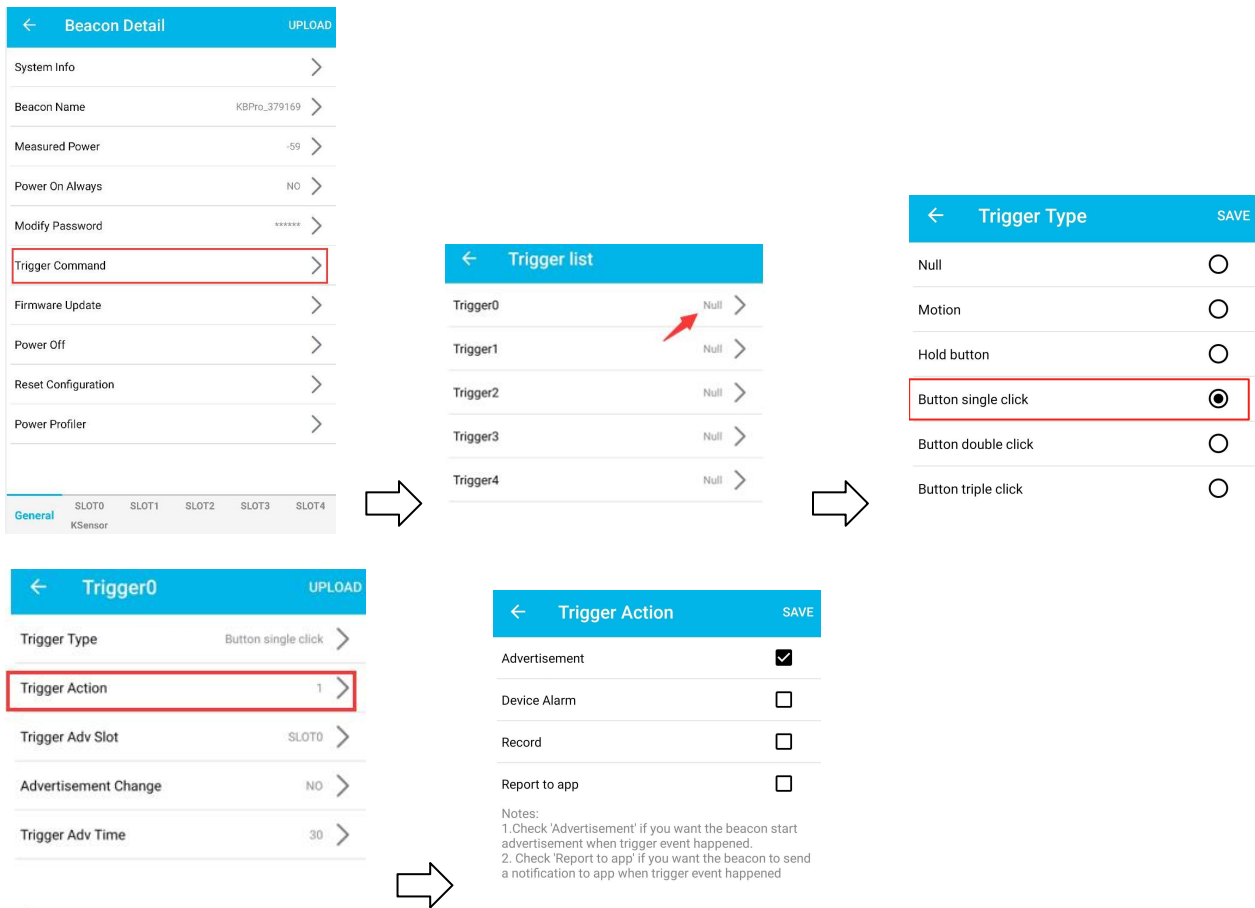
4. How to Configure Trigger

For some KBeacon device that has some motion sensor, temperature&humidity sensor, push button, etc., The application can configure the KBeacon to monitor some trigger event. For example, button was pressed, the temperature is too high, or device was motion. The KBeacon can do some action when the trigger condition was met.

When the conditions of the Trigger are met, we can trigger a specific advertisement, or trigger the event to be recorded in memory, or report an event to the App.

Tap: Trigger Command—>Trigger Type

The following example is based on the K21 device



4.1 Trigger event to advertisement

The trigger advertisement has follow parameters:

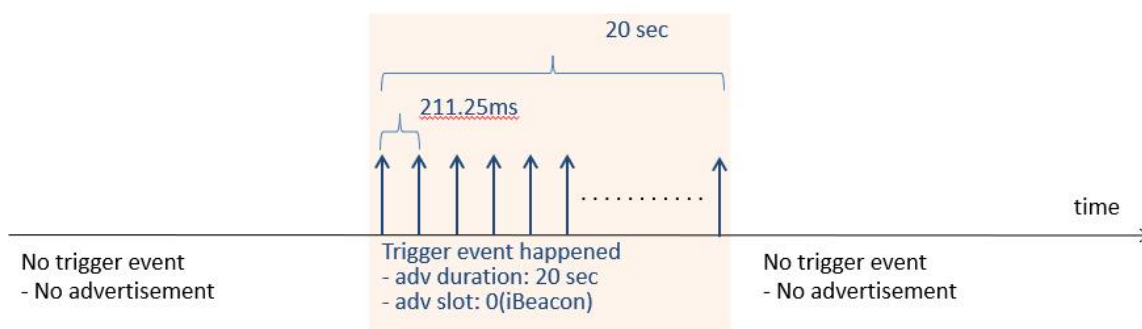
- Trigger No: Trigger instance number, the device supports up to 5 Triggers by default, the No is 0 ~ 4.
- Trigger type: Trigger event type
- Trigger action: Action when trigger event happened. For example: start broadcast, make a sound, or send a notification to the connected App.
- Trigger Adv slot: When the Trigger event happened, which advertisement Slot starts to broadcasting
- Trigger parameters: For motion trigger, the parameter is acceleration sensitivity. For temperature above trigger, you can set to the temperature threshold.
- Trigger Adv duration: The advertisement duration when trigger event happened. Unit is second.
- Trigger Adv TX power: The advertisement TX power when trigger event happened. Unit is dBm.
- Trigger Adv interval: The advertisement interval when trigger event happened. Unit is ms.

4.1.1 Trigger only advertisement

The device usually does not broadcast by default, and we want to trigger the broadcast when the trigger event happened.

Example:

1. Setting slot 0 to iBeacon advertisement(adv period = 211.25ms, trigger only adv = true).
2. Add a single button trigger(Trigger No = 0, Trigger type = Btn single click, Action = advertisement, Adv slot = 0, Adv duration = 20).



4.1.2 Trigger to an exist advertisement

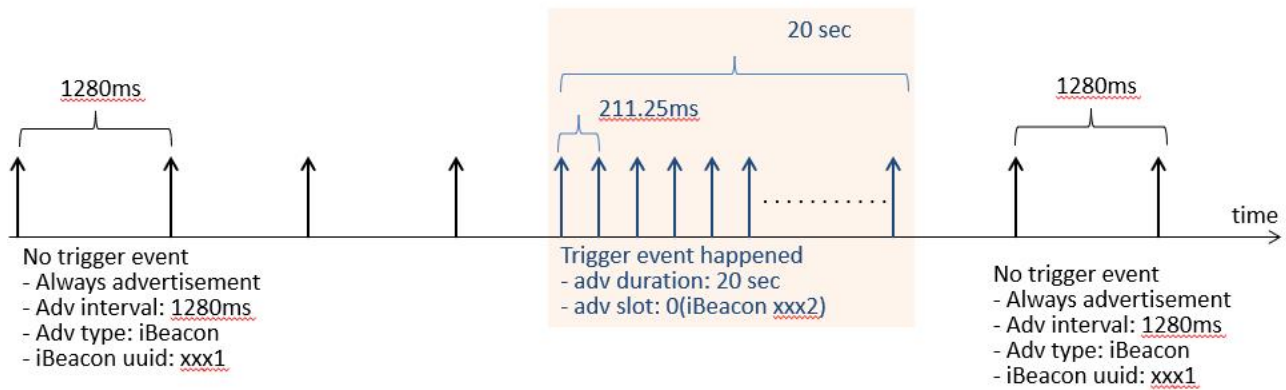
For some scenario, we need to continuously monitor the KBeacon to ensure that the device was alive. The device usually broadcasting iBeacon1 (UUID=xxx1) , and we want to trigger the broadcast iBeacon2 (UUID=xxx2) when the button is pressed.

Example:

1. Setting slot 0 to iBeacon advertisement (UUID=xxx1, adv period = 1280ms, trigger only adv = false).
2. Setting slot 1 to iBeacon advertisement (UUID=xxx2, adv period = 211.25ms, trigger only adv = true).

We set an larger advertisement interval during alive advertisement and a short advertisement interval when trigger event happened, so we can achieve a balance between power consumption and triggers advertisement be easily detected.

3. Add a single button trigger(Trigger No = 0, Trigger type = Btn single click, Action = advertisement, Adv slot = 1, Adv duration = 20).



4.1 Trigger event to record

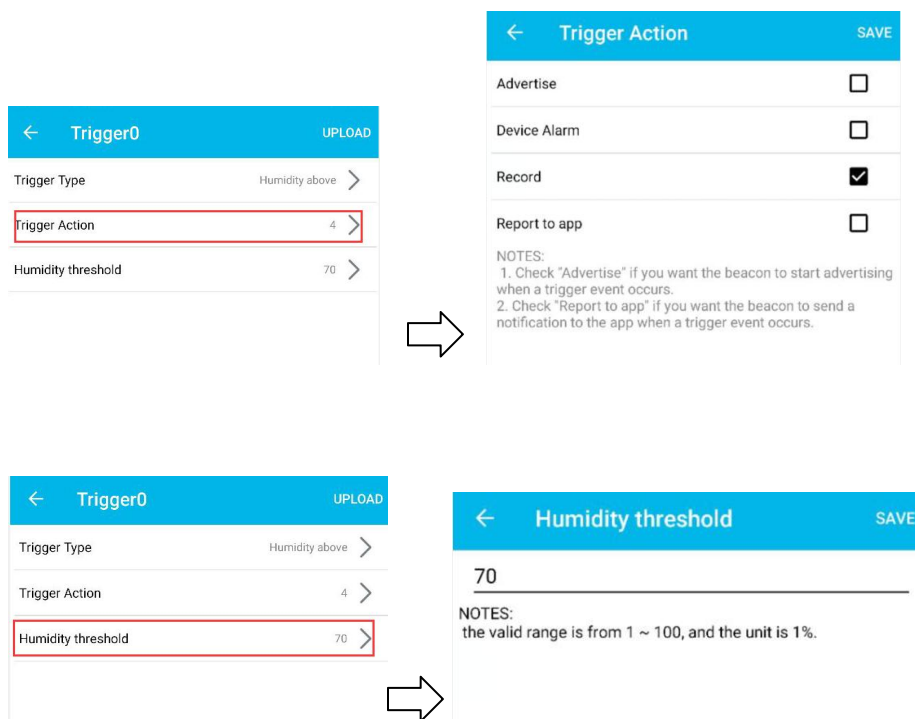
For some Triggers, the device supports recording the Trigger events. For more information, see 2.1.1 Trigger capability.

For the Trigger event, we can set the Trigger Action to "Record". After setting, when the trigger event is triggered, KBeacon will record the Trigger event.

Reminder: Currently, only some devices support recording Trigger events, including:

Example:

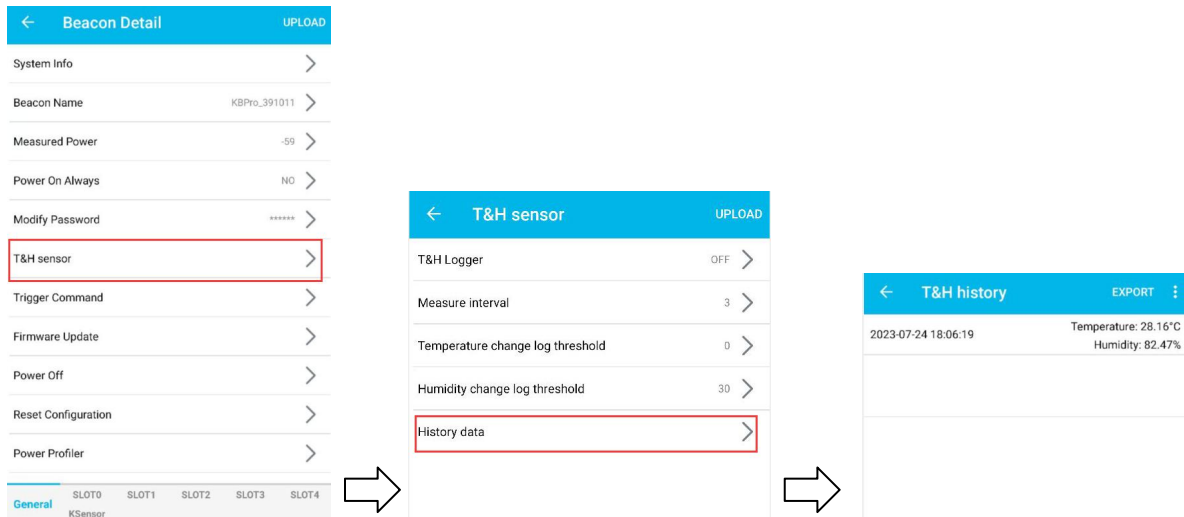
1. We assume that the current ambient humidity is 60%, and we set an trigger event to be logged when the humidity exceeds 70%.



2. Put the KBeacon in an environment with a humidity over 70%.

3. Observe whether the event is logged.

In order to verify that the record is generate by the Trigger, we can turn off the T&H Logger during the test, which means that when the temperature and humidity change exceeds the specified threshold, the recording will not be performed. It will only be logged when the Trigger event happened.



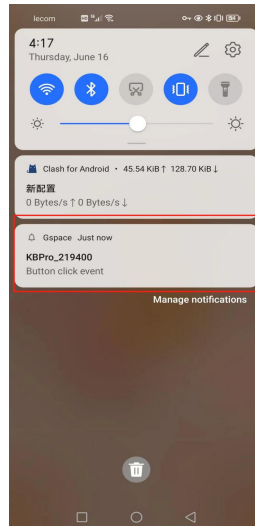
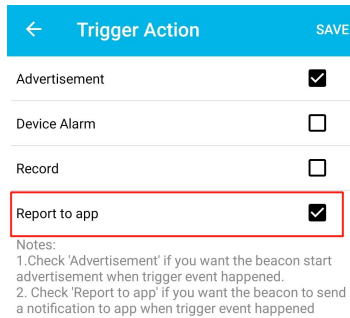
4.2 Trigger event to App

We can also set KBeacon to send an event message to the App when the Trigger is happened.

Trigger Action	SAVE
Advertisement	☒
Device Alarm	☐
Record	☐
Report to app	☐

Notes:
 1. Check 'Advertisement' if you want the beacon start advertisement when trigger event happened.
 2. Check 'Report to app' if you want the beacon to send a notification to app when trigger event happened

If set “Report to App”, there will be a message notification displayed on the cell phone when the trigger event happens.



5. How to batch configure Beacon

For KBeacon, we also offer a batch configuration feature. We do the batch configuration of Beacon through another app. Through the batch configuration application, you can set the parameters that need to be configured in advance, then add the beacon that needs to be configured, and then start the configuration. The Application will automatically configure the Beacon devices one by one.

The name of the APP is KBProBatchCFG, you can download it from:

5.1 Download KBProBatchCFG App



iOS APP



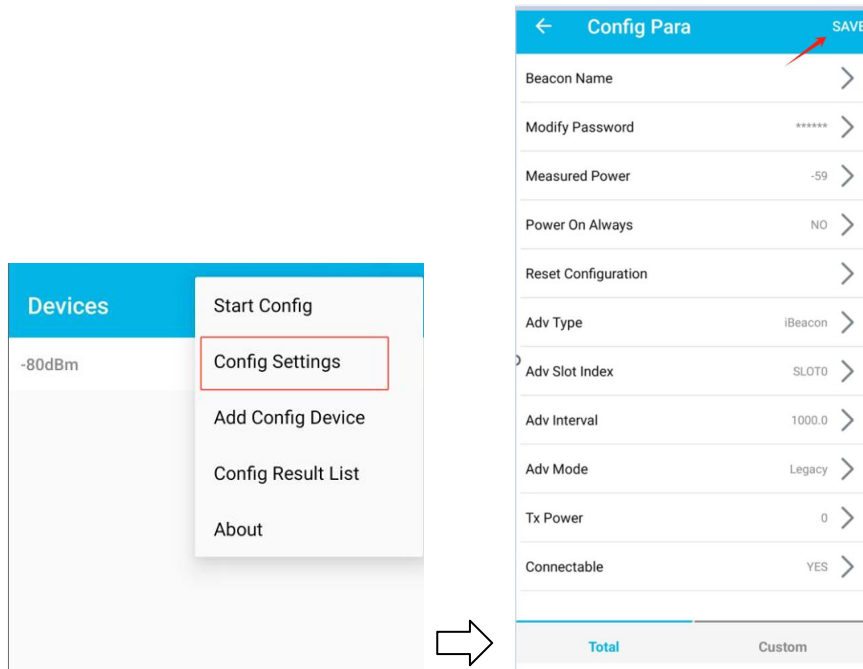
Android APP

5.2 Batch configure steps

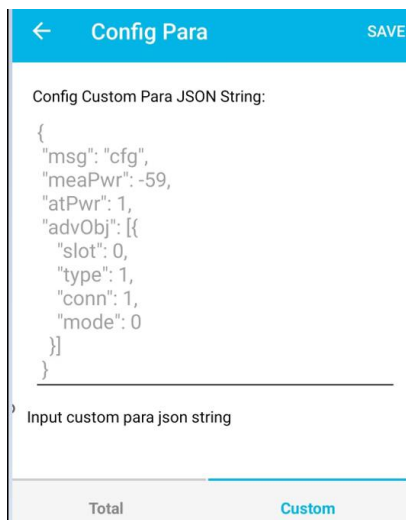
Step 1: Create configuration parameters

Tap: Config Settings—> Config Para—>Save.

The following screenshot shows the contents of the configurable parameters, only one slot can be configured at a time. When you're done configuring the parameters, click Save.



If the parameter you want to configure is not in the following screenshot, you can configure it by inputting the custom para json string, please contact KKM sales for more information.



Step 2: Add Configuration Devices

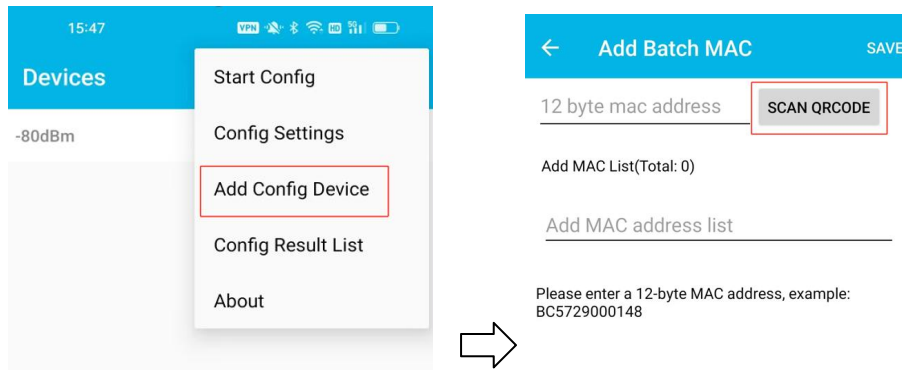
The batch configuration application supports adding KBeacon by scanning the QR Code. The tool also supports adding Beacons via Bluetooth scanning.

(Reminder: 1. This step is different between Android and IOS, Android supports both ways. The IOS can only add the device list by scanning.

2. The APP will no longer scan devices that have been configured. Before adding a device, please clear the history of the last configuration. Configure the results list via the menu and tap Clear.)

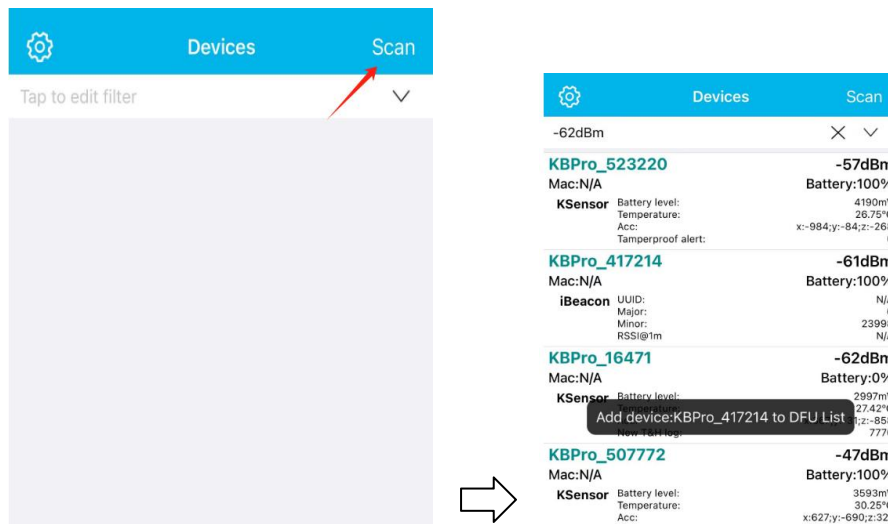
a) Add device by scan QRCODE

Tap: Add Config Device—> SCAN QRCODE—> Save



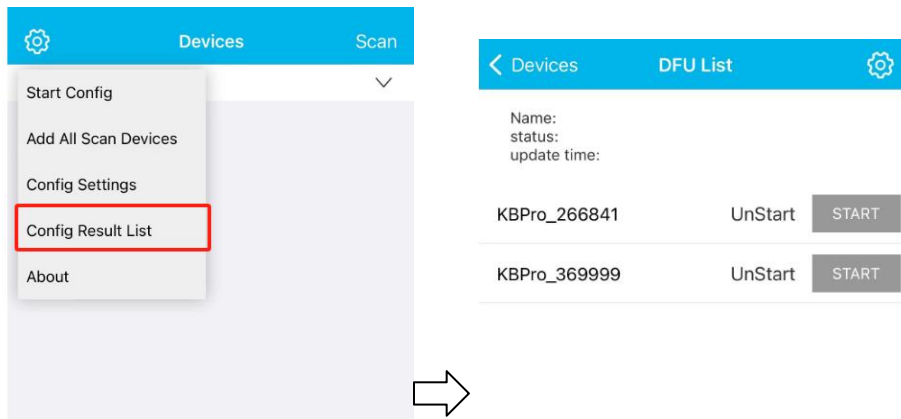
b) Add device by BLE scanning

Tap on Scan to start BLE scanning. You can also set the filtering parameters before starting the scan.



View the list of devices you added

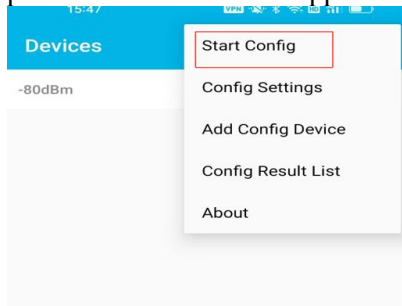
Tap on Config Result List, You can view the added devices. UnStart means batch configuration is not start.



Step 3: Start Batch Configuration

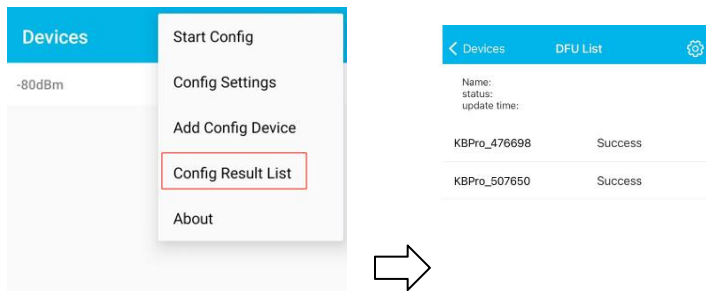
Tap on “Start Config” will start batch configuration.

During batch configuration, please keep the Android/iOS Application in the foreground. Don't turn off your phone screen. When the app is in the background, the system may not allow the application to execute.



Step 4 View Configuration Result

After configuration complete, you can view the results by clicking Config Result List.



5.3 Batch Configuration FAQ

1. Batch Configuration Tool can't scan to the Beacon device?

The KBProBatchCFG APP will not scan the devices that are already in the configuration list when scanning, this may happen because you have already added the device or configured the device before. Therefore, you need to clear the previous configuration record before you can add it. Through the menu Config Result List => Clear

2. The KBProBatchCFG APP can't connect to the device after a period of time in batch configuration?

The app is running, please keep the screen always lit. In addition, please set the app to run in the foreground.

3. I need to configure Trigger and Sensor parameters, the current batch configuration does not support input? For other parameters such as Trigger and Sensor, our batch configuration tool supports configuration via JSON messages. Please tell us the Trigger or Sensor parameters you need to configure and contact our sales to help you generate JSON configuration parameters.

4. Why the IOS KBProBatchCFG APP does not support to add the devices to be configured by scanning the QR code?

Because the iPhone OS doesn't allow the APP to get the MAC address of the device when scanning the Beacon device. Therefore, the APP can't add the device to be configured by scanning the MAC address of the QR code. You can only add the devices to be configured by scanning the Beacon devices then add the devices to the configuration list and then start the configuration.

6. Other Settings

6.1 Unconnectable mode

Each SLOT has two different advertising status, connectable mode and unconnectable mode. Only when the advertising status is connectable, the KBeacon is configurable. But the unconnectable mode saves about 20% -30% power consumption than connectable mode.

After the KBeacon is deployed, we recommend setting the KBeacon to be unconnectable mode. This can lower battery power consumption and the Beacon also have better security performance.

How to set unconnectable mode:

Tap: Connectable—> OFF—> Save—>Return —>UPLOAD.

Beacon Detail		UPLOAD
Beacon Type	iBeacon	>
Adv Interval	1000.0	>
Tx Power	0	>
Adv Mode	Legacy	>
Connectable	YES	>
Trigger only advertisement	NO	>
UUID	7777772E-6B6B-6D63-6E2E-636F6D000001	>
MajorID	1	>
MinorID	1	>
General SLOTO iBeacon SLOTT SLOTT SLOTT SLOTT		



Connectable		SAVE
NO	☐	
YES	☒	
NOTES: 1: Setting Connectable to NO can prevent other devices from maliciously connecting after your beacon is deployed. In addition, setting the beacon to non-connectable mode can reduce power consumption by about 20 ~ 30%. 2: If you want to connect the device again, you can do the following two ways: a): Single click the button of the device, the device will enter the connectable mode for 30 seconds. b): Reinstall the battery, the device will enter the connectable mode for 30 seconds. WARNING: If your device does not have a button and cannot be opened to re-install the battery, once your device is in non-connectable mode, you will no longer be able to connect the device again.		

Warning: If your device is in the following two situations, once you set No connectable, you will not be able to connect the device again:

1. The device does not have a button and cannot be opened to reinstall the battery. For example, K12, T1, etc.
2. The device has a button and a Button Trigger was set, and it cannot be opened to reinstall the battery. For example, K7, F1, etc.

Question: How can I configure KBeacon again if it was set to be unconnectable mode?

- For KBeacon with button: click the button, the Beacon will enter a connectable mode for 30 seconds, users can connect the device within these 30s. Or re-install battery.

(Reminder: If the KBeacon device is configured with button trigger, the Beacon will NOT enter a connectable mode by clicking the button.)

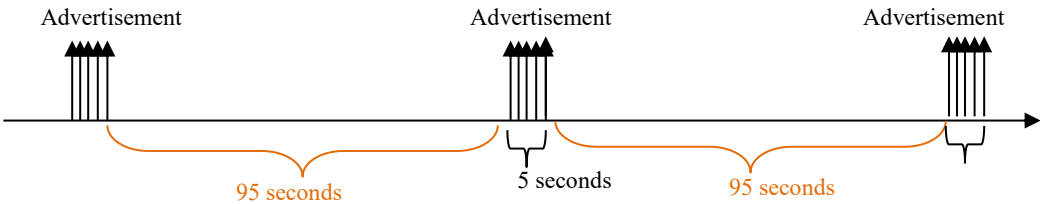
- For KBeacon without button: Re-install battery

(Reminder: For the KBeacon device that doesn't have button and whose battery can not be re-installed, once the device was set to unconnectable mode, it can not be configured any more!)

6.2 Periodically advertisement

In some scenarios, we need Beacon to start broadcasting at a certain duration and then sleep.

For example, we need Beacon to broadcast for only 5 seconds every 100 seconds. That is to say, Beacon sleeps from 1 to 95 seconds, and broadcasts from 96 to 100 seconds with interval of 1 second.



Configuration:

Step1: Set SLOT0 to Trigger Only, which means that it does not broadcast by default and can only be broadcast when triggered by a Trigger.

←

Beacon Detail

UPLOAD

Adv Type

iBeacon

>

Adv Interval

1000.0

>

Adv Channel

7

>

Tx Power

4

>

Connectable

YES

>

Trigger Only Adv

YES

>

UUID

7777772E-6B6B-6D63-6E2E-636F6D000001

>

Major ID

1

>

Minor ID

1

>

General

SLOT0

SLOT1

SLOT2

SLOT3

SLOT4

iBeacon

Step2: Add a periodic Trigger with an interval of 100 seconds. When an event occurs, trigger the Beacon to broadcast for 5 seconds with an interval of 1 second.

← Beacon Detail		UPLOAD
System Info		>
Beacon Name	KBPro_476719	>
Measured Power	-59	>
System Info		>
Power On Always	NO	>
Modify Password	*****	>
Trigger List		>
Firmware Update		>
Power Off		>
Reset Configuration		>
Power Profiler		>

← Trigger Type		SAVE
Null	☐	
Hold button	☐	
Single click	☐	
Double click	☐	
Triple click	☐	
Periodically Event	☒	

← Trigger 0		UPLOAD
Trigger Type	Periodically Event	>
Trigger Action	1	>
Trigger Adv Slot	SLOT0	>
Advertisement Change	NO	>
Trigger Adv Duration	5	>
Trigger Adv Interval	1000.0	>
Trigger Adv Power	0	>
Event Interval	100000ms	>

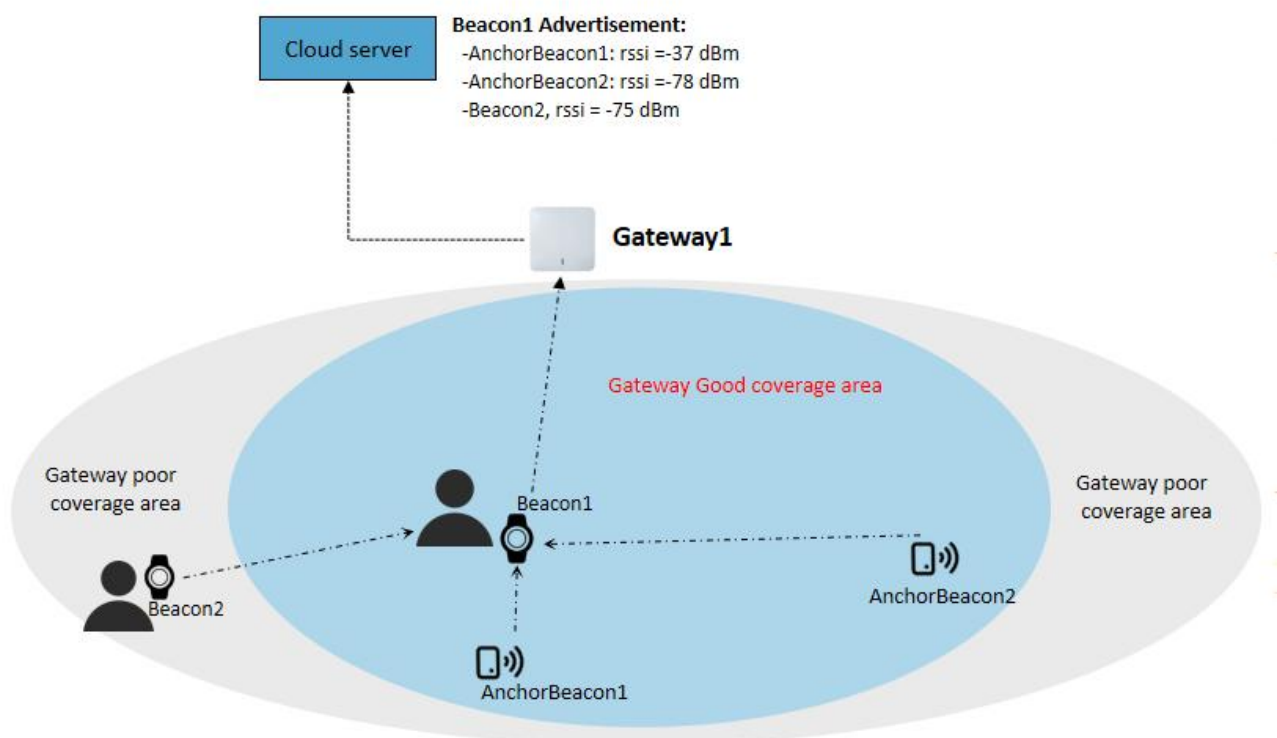
6.3 Repeater Beacon

Currently, the typical iBeacon positioning solution works as follows: the Beacon broadcasts signals, and the Gateway scans the Beacon signal strength. Positioning is achieved using the RSSI of the scanned Beacon. To improve positioning accuracy and ensure that Beacons in weak signal coverage areas are scanned, the usual method is to increase the number of Gateways. By capturing the RSSI of a single Beacon with multiple Gateways and applying the triangulation principle, the position of the Beacon can be calculated. Theoretically, the more Gateways deployed, the wider the coverage and the higher the positioning accuracy. However, the deployment of Gateways

becomes more challenging as they require power supply and network connectivity.

KKM's Repeater solution effectively addresses these issues. The Repeater solution allows a Beacon to not only broadcast iBeacon signals but also scan nearby Beacon signals. When nearby Beacon signals are detected, the Repeater can send the IDs of these Beacons to the Gateway by including them in its broadcast messages. This solution offers the following advantages:

- **Extending the positioning range of Beacons:** In weak indoor coverage areas, some Beacons may not be scanned by the Gateway. Other Beacons will act as Repeaters and relay the signal of those Beacons to the Gateway. The Gateway can then position the Beacons in weak coverage areas.
- **Improving positioning accuracy:** Multiple Anchor Beacons can be deployed in fixed indoor locations. A mobile Beacon can periodically scan the Anchor Beacons and send the scanned Anchor Beacon information to the Gateway. The Gateway can then accurately position the mobile Beacon based on the Anchor Beacon information



At present, KKM Repeater supports the following two ways to trigger scanning:

1. Periodically scanning of peripheral devices

Beacon periodically scans peripheral devices and broadcasts the MAC addresses of the scanned peripheral devices through iBeacon advertisement.

2. Motion trigger scanning of peripheral devices

Beacon start scanning after motion trigger, and then broadcasts the scanned device MAC address and RSSI through iBeacon advertisement.

The broadcast format of the scanned devices carried in iBeacon is as follows:

iBeacon	UUID(16 Byte)				Major	Minor	Power
iBeacon + Scanning Device	MAC1(3Byte)+RSSI(1byte)	MAC2+RSSI	MAC3+RSSI	MAC4+RSSI	Major	Minor	Trigger Event

The Repeater solution use UUIDs to carry the MAC address and RSSI information of scanned devices. The UUID can carry up to 4 scanned beacons, and each beacon occupies 4 Bytes. The MAC address is only 3 bytes, so we omitted the 3 high-Bytes of the MAC address. For example: If MAC1 address is BC5729010417 and the RSSI is -78dBm, then the value is 010417B2.

If the number of devices scanned beacon is less than 4, the that part of the UUID will be filled with 0. If more than 4 devices are scanned, the Beacon will fill the UUID with the 4 strongest RSSI signal Beacons.

MARK: The repeater solution only supports scanning KKM's Beacon. All KKM beacons start with BC5729.

Example:

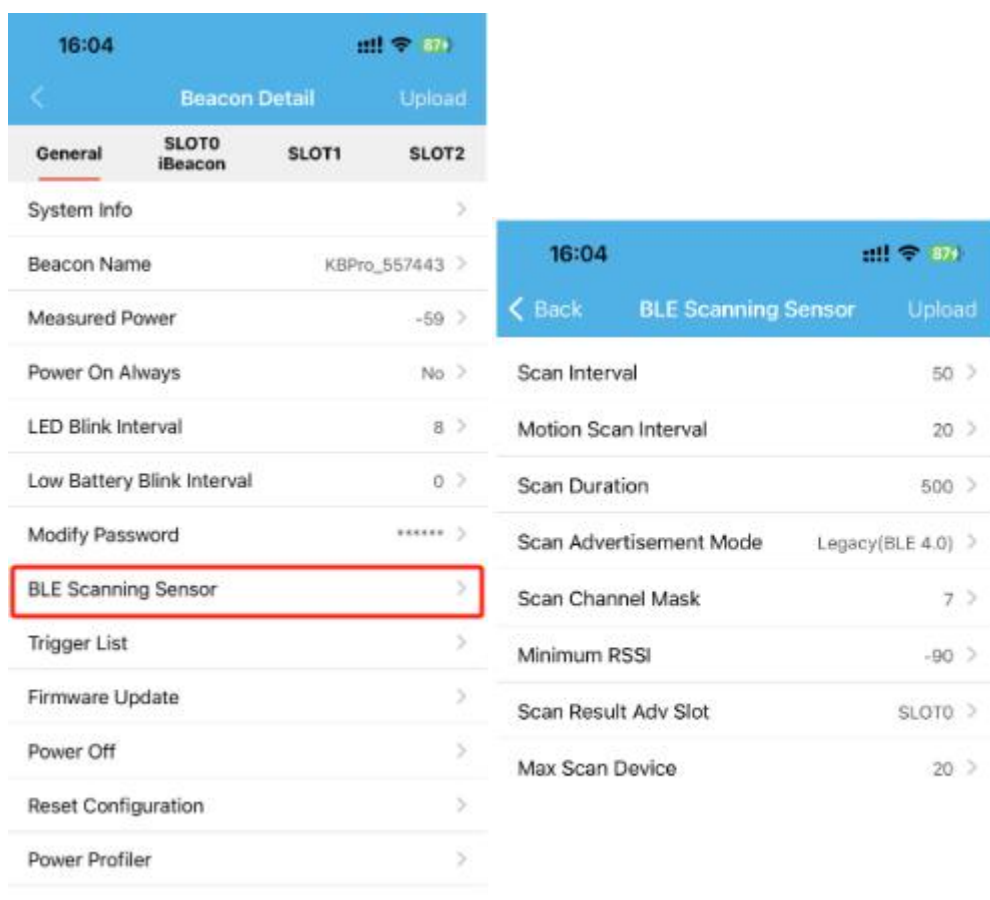
1. Assuming SLOT0 is configured as iBeacon, the UUID is 1332F93C-A3CD-E044-4599-A9C2D462E557, the Major is 12, the minor is 14, and the reference power is -59dBm
2. Assuming the scanning is triggered by periodic triggers(trigger event value is 20)
3. Assuming Beacon scans 5 devices, namely:
 - BC5729041719, rssi: -78
 - BC572904171A, rssi: -71
 - BC572904171B, rssi: -68
 - BC572904171F, rssi: -87 (The signal is the worst and will not be included in UUID)
 - BC572904171C, rssi: -69

When Trigger is triggered, the broadcast information is as follows

iBeacon	UUID(16 Byte)				Major	Minor	Power
Scanning Data	0x041719B2	0x04171AB9	0x04171BBC	0x04171CBD	0x000C	0x000E	0x14

Note: This feature indicates that only some beacons can support it. Please contact sales to confirm which models support this feature.

6.3.1 How to Set BLE Scanning Event



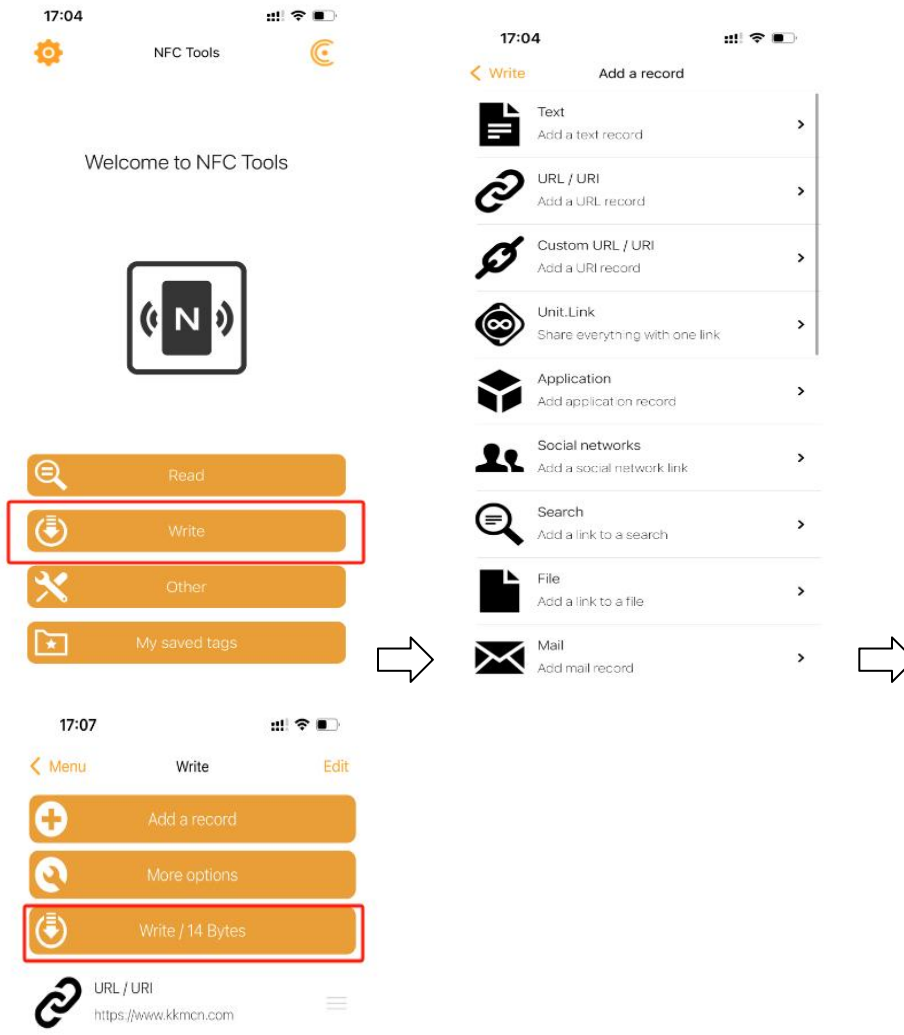
Connect the KBeacon which has Repeater functionality to the KBeaconPro APP, and click on the BLE Scanning Sensor on the General page to configure the relevant parameters. Below are the explanations for the related parameters:

- **Scan Interval:** Defines the interval at which scanning events occur. Setting this to 0 disables the feature.
- **Motion Scan Interval:** Once motion is detected, this scanning event is triggered. The device remains in a motion state for up to 2 minutes. If no further motion is detected within this period, the scanning interval reverts to the **Scan Interval**. Setting this to 0 disables the feature. The unit is in seconds.
- **Scan Duration:** The duration of each scanning action.
- **Scan Advertisement Mode:** This indicates the type of broadcast to scan for.
- **Scan Channel Mask:** This specifies which channel's broadcast data to scan.
- **Minimum RSSI:** If the RSSI value of a scanned BLE device is below this threshold, that device will be ignored.
- **Scan Result Adv Slot:** A specific slot allocated for handling the outcomes of scanning activities.
- **Max Scan Device:** If the number of scanned devices exceeds this parameter, scanning will stop even if the Scan Duration limit has not been reached.

MARK: The Event Interval time > BLE Scan Duration + Trigger Adv Duration

6.4 NFC configuration

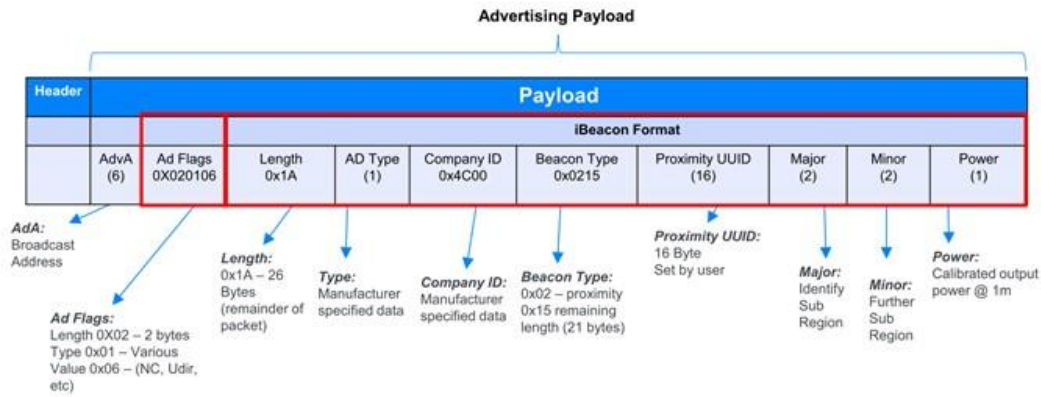
Products with NFC capabilities, such as K7, F1, etc., can be read and written using third-party NFC tools, such as Email, URL, Text, etc. The following tool is a third-party app NFC Tool.



7. Advertisement and scan response payload

7.1 iBeacon advertisement payload struct

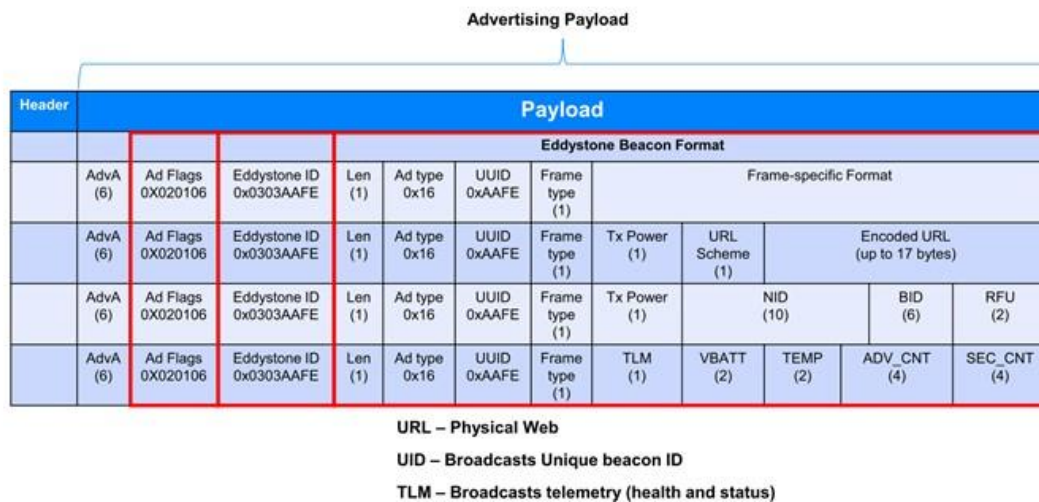
iBeacon is a BLE broadcast format defined by Apple for Location. For detailed definitions see: <https://developer.apple.com/ibeacon/>



7.2 Eddystone advertisement payload struct

Eddystone is a BLE broadcast format defined by Google. For detailed definitions see:

<https://github.com/google/eddystone>



7.3 KSensor advertisement payload struct

The advertising data packet of KSensor is defined by KKM. KSensor is used to broadcast some sensor data, such as temperature and humidity, light intensity, and so on. Please refer to the following table for details.

Length	Type	value	Remark
Head	AdvA	0x6	
	(1byte)		
	AdvFlags	0x020106	
	(3byte)		
	services ID	0x0303AAFE	

	(4byte)		
	Length	Variable, the length of all subsequent messages	Services data length.
	(1byte)		
	Adv type	0x16	
	(1byte)		
	Services UUID	0xAAFE	
	(2byte)		
	Frame Type	0x21	
	(1byte)		
Sensor Mask	Sensor mask	0bit: voltage indication; 1bit: temp indication 2bit: humidity indication 3bit: acc indication 4bit: cutoff indication 5bit: PIR indication 6bit: Light indication 7bit: VOC indication 8bit: reserved; 9bit: CO2 indication 10bit: Unread record count 11~15bit: reserved	
Voltage	Voltage	Battery voltage, unit is mV	
	(2byte)		
Temperature	Temperature	Temperature, Fixed Point 8.8 format	
	(2byte)		
Humidity	Humidity	Temperature, Fixed Point 8.8 format	If the humidity indicator bit is 1, this field is valid, otherwise it is empty
	(2byte)		
Acc axis	Axis X pos	Axis X, unit is mg	If the acc indicator bit is 1, this field is valid, otherwise it is empty
	(2byte)		
	Axis Y pos	Axis Y, unit is mg	
	(2byte)		

	Axis Z pos (2byte)	Axis Z, unit is mg	
Alarm	Cutoff alert (1byte)	For S1: ● bit0: 1 : door open, 0 closed ● bit1: 1: device was plug, 0 unplug For W3/W7: ● bit0: 1 watchband alarm, 0 normal For S4: ● bit2: 1: detected water, 0, no detected water For S7: ● bit0: 1 parking was detected, 0 no parking ● bit5: 1:device dose not have idle parking reference value ● bit6: 1:device's geomagnetic sensor not calibrated	If the cutoff indicator bit is 1, this field is valid, otherwise it is empty
PIR	PIR alert (1byte)	bit0: 1 PIR alert, 0 no alert	If the PIR indicator bit is 1, this field is valid, otherwise it is empty
Light	Light level (2byte)	Light level. Unit is lux	If the Light indicator bit is 1, this field is valid, otherwise it is empty
Unread record count	record type (1byte)	0bit: indicate follow two byte contain temperature and humidity record count	
	record count (2byte)	Unread record count	

Remark:

- The unit of battery is mV. For example, if the VBATT is 3270, it means the battery voltage is 3270mV;
- Temperature. 2 bytes Fixed Point 8.8. The format is same as temperature in Eddystone TLM.
- Humidity. 2 bytes Fixed Point 8.8. The format is same as temperature in Eddystone TLM.

The Fixed Point 8.8 format:

<http://people.ece.cornell.edu/land/courses/ece4760/>

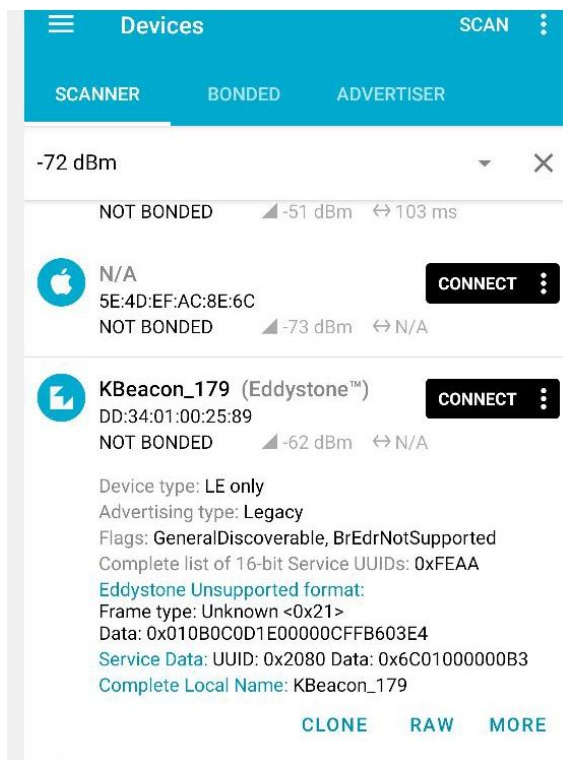
Python example:

```
def get_f16(temperature):
    if temperature < 0x8000:
        temperature = temperature
    else:
        temperature = temperature - 0x10000
    return round((temperature / 256.0), 2)

print(get_f16(0x1D00))
```

Example :

View KSensor data by third part NRF connect application(Android version)

**7.4 EBeacon(Encrypt UUID) advertisement payload struct**

Offset	Length	Type	value
0	1 byte	AdvA	0x6
1	3 byte	AdvFlags	0x020106
4	1 byte	Length	0x1A
5	1 byte	Type	0xFF
6	2 byte	Manufacture ID	0xF530

8	1byte	Beacon Type	0x03: MD5 + AES ECB 0x05: SHA256 + AES CBC
9	1byte	Remain Length	0x15
10	16byte	UUID	Encrypted UUID
26	4byte	UTC	UTC seconds, big endian
30	1byte	Measure power	Default is -59@1m

For detailed encryption introduction, please contact our sales by sales@kkmcn.com

7.5 System advertisement payload format

The advertising data packet of System is defined by KKM. The broadcast of the System type is mainly used to broadcast the MAC address of the device and some system parameters of the device, such as the device model and battery level.

Since the IOS system does not allow the application to obtain the MAC address of the BLE device, through System broadcast, the APP can obtain the MAC address of the device without connecting to the device.

Please refer to the following table for details.

Offset	Length	Type	value
0	1byte	AdvA	0x6
1	3byte	AdvFlags	0x020106
4	4byte	ID	0x0303AAFE
8	1byte	Length	
9	1byte	Adv type	0x16
10	2byte	UUID	0xAAFE
12	1byte	Frame Type	0x22
13	1byte	Model ID	Device model ID
14	1byte	Battery	Battery Percent
15	6byte	Mac address	big-endian
21	2byte	Software version	big-endian format. For example if the value is 0x0632, then version is V6.49

Model ID define:

- K1: 10
- K11: 11
- K12: 12
- B1: 13
- U1: 14
- F1: 15

- K15a: 16
- K15:19
- K18: 18
- S1: 17

- K21: 21
- B2: 22
- K21u:25
- S2:26
- K23: 23
- K23p: 24
- F2: 27
- S23:28
- P2es:29

- K3: 30
- W3: 38
- S3: 36
- C3: 31
- C3p:37
- C3ps:36

- K4: 40
- K41:41
- K4u: 45
- K4p: 47
- W4: 48
- K4pt: 49
- S4: 46
- k41ph: 43

- K5: 50
- K51: 51
- W52: 52
- K5p: 57
- K5pt: 58
- S5:59
- S51: 56
- B5: 55
- W59u: 53
- W59:54

- K6: 60
- K61: 61
- K6p: 67
- K6pb: 68
- K6ps: 69
- S6: 64
- S6e:62
- S6t: 65

- K7: 70
- K71: 71
- K7u: 75
- W7: 78
- M7:73
- S7: 74

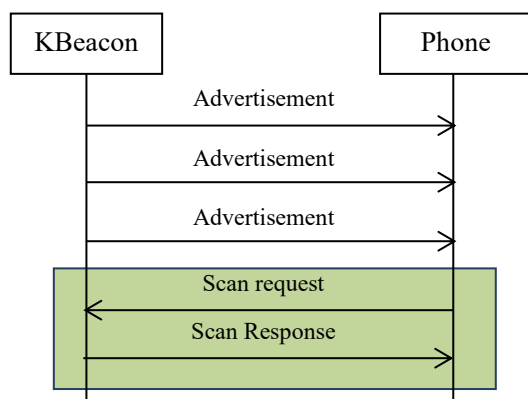
- K8: 80
- K8u: 85
- W8:86
- W8u:87
- W81:88
- S8:89

- K9: 90
- K91: 91
- K9P: 97
- K9Pb: 98
- K9Pu:99
- W9:92
- K9ps:93

- P1:101
- B1u:102
- C2: 102
- P2:103
- K16a:104
- K16:105
- K17:106

- T1:107
- T1a:108
- P11:109
- K7pa: 110
- K5pa: 111
- W1:112
- F1s:113

7.6 Scan response payload format



1. KKM services data. it include battery level, iBeacon major id, iBeacon minor id. These data was included in service ID(00002080-0000-1000-8000-00805f9b34fb)

Length	Type	value	Remark
1byte	Data length	0x9	Service data length
1byte	Data tye	0x16	Service data type
2byte	Service ID	0x8020	
1byte	Battery percent		If value > 100, the battery is 100%
1byte	Adv type	0x1	
2byte	MajorID		
2byte	MinorID		

2. device name.

Device name follows standard Bluetooth protocol.

Length	Type	value	Remark
1byte	Data length	variable	Device data length
1byte	Data tye	0x9	Full device name
Variable	Device data length		Ascii device name